



1. RESEARCH QUESTION

- Q1** Does current cigarette use differ between male and female students?
- Q2** Are female smokers behaviorally different from female non-smokers?
- Q3** Do behavioral risk characteristics become more prevalent as smoking intensity increases?

2. DATA

Dataset: YRBS 2007

Sample Size

Total Students	14,041
Cleaned Sample	13,312
Female Students	7,036

Key Variables



• CurrentCigaretteUse

Behavioral Factors

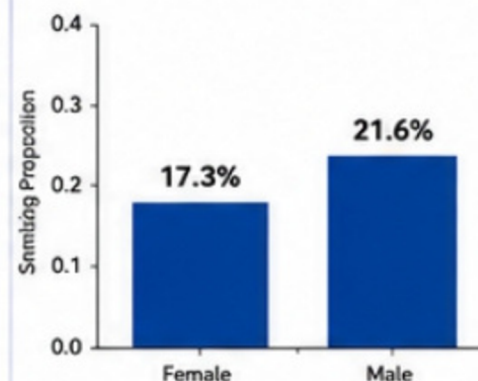
- Alcohol Use
- Binge Drinking
- Marijuana Use
- Physical Fighting
- Sleep
- SadOrHopeless

3. EDA HIGHLIGHTS

Gender Difference (Notebook 1)

Male students reported higher smoking prevalence.

Current Cigarette Use Proportion by Gender



Difference = 4.26%
(95% CI: 0.029 – 0.056)
 $z = 6.215, p < 0.001$

Female Smoker Profile (Notebook 2)

Compared with female non-smokers, smokers showed higher risk in most behaviors and shorter sleep duration.

Behavioral Profile of Female Smokers and Non-Smokers

	Alcohol Use	Binge Drinking	Marijuana Use	Physical Fighting	Sleep
Non-Smoker	1.55	1.28	1.16	1.44	3.83
Smoker	2.90	2.44	2.22	1.99	3.39

All Mann-Whitney U tests $p < 0.001$
SadOrHopeless: $\chi^2 = 214.61, p < 0.001$

Smoking Intensity Trend (Notebook 3)

Risk behaviors generally increased as smoking intensity increased.

Behavioral Risk Trends Across Smoking Intensity Groups

	Alcohol Use	Binge Drinking	Marijuana Use	Physical Fighting	Sleep
Non-Smoker	1.55	1.28	1.16	1.44	3.83
Light	2.76	2.36	1.86	1.83	3.56
Moderate	3.32	2.68	2.24	2.22	3.32
Heavy	3.24	2.84	3.06	2.08	3.07

All Chi-square tests $p < 0.001$
Strongest association: Marijuana Use ($\chi^2 = 1981.21$)

4. METHODS

Notebook 1

Two-Proportion Z-Test

- Compare smoking proportion between Male and Female

Notebook 2

Mann-Whitney U Test

- Compare female smokers vs. female non-smokers

Notebook 3

Chi-Square Test of Independence

- Compare smoking intensity groups (Non-Smoker, Light, Moderate, Heavy)

5. KEY RESULTS

A. Gender Difference (Notebook 1)

Female Proportion	17.3%
Male Proportion	21.6%
Difference (Male – Female)	4.26%
95% Confidence Interval	(0.029, 0.056)
z Statistic	6.215
p-value	< 0.001

Conclusion: Reject H_0 . Male students have a higher smoking proportion.

B. Female Smoker Profile (Notebook 2)

All behavioral variables (Mann-Whitney U tests)

Alcohol Use	$p < 0.001$	↑ Alcohol Use
Binge Drinking	$p < 0.001$	↑ Binge Drinking
Marijuana Use	$p < 0.001$	↑ Marijuana Use
Physical Fighting	$p < 0.001$	↑ Physical Fighting
Sleep (lower is shorter)	$p < 0.001$	↓ SadOrHopeless
		↓ Sleep Duration

SadOrHopeless (Yes %)
 $\chi^2 = 214.61, p < 0.001$

C. Smoking Intensity Trend (Notebook 3)

All Chi-square tests $p < 0.001$

Alcohol Use	$\chi^2 = 1720.87$
Binge Drinking	$\chi^2 = 1580.17$
Marijuana Use	$\chi^2 = 1981.21$
Physical Fighting	$\chi^2 = 298.78$
SadOrHopeless	$\chi^2 = 190.99$

SadOrHopeless (% Yes)

Smoking Intensity	% Yes
Non-Smoker	32.82%
Light	53.15%
Moderate	56.72%
Heavy	54.25%

Risk behaviors generally increase with smoking intensity.
Strongest upward trend: Marijuana Use. Sleep shows the opposite trend.

6. CONCLUSION (KEY TAKEAWAY)

Gender Difference

Male students showed a significantly higher prevalence of current cigarette use than female students.

Female Smoker Profile

Female smokers exhibited a broader behavioral and psychological risk profile and reported shorter sleep duration.

Smoking Intensity Trend

Behavioral risk characteristics generally became more prevalent as smoking intensity increased, while sleep duration decreased.

Overall Conclusion

Smoking among female students is part of a broader behavioral risk profile. These findings support targeted interventions addressing multiple co-occurring risk behaviors in adolescents.